

SYLVIA GAO

(617) 706-1136 | sylviagao00@gmail.com | [LinkedIn](#) | Boston, MA (Willing to Relocate)

EDUCATION

Northeastern University

M.S. in Analytics | GPA 3.9

Boston, MA

May 2025

Tianjin Polytechnic University

B.A. in Marketing | GPA 3.5

Tianjin, China

May 2022

SKILLS

Programming: Python (pandas, NumPy, matplotlib, seaborn, scikit-learn), R, SQL (PostgreSQL)

Database & Cloud: AWS (Glue, Redshift, S3, Lambda), Hive, Spark

BI & Visualization: Power BI, Tableau, Looker, Qlik, Google Analytics, Metabase, Excel (Vlookup)

Miscellaneous: A/B testing, Data Modeling, Clustering, Predictive Modeling, ad-hoc, Risk Management

PROFESSIONAL EXPERIENCE

Buyer Folio (Capstone Project)

Boston, MA

Data Analyst

Jan. 2025 – Apr. 2025

- Optimized real estate market value forecasting models, used to guide resource allocation by enhancing XGBoost feature selection and hyperparameter tuning; improved model precision from 78% to 83%, boosting ROI
- Automated real-time data integration for property price analytics; engineered ETL pipeline connecting PostgreSQL to AWS Redshift using Python; optimized database indexing, reducing report generation time by 65%
- Developed Tableau interactive dashboard incorporating K-means clustering and hierarchical analysis; applied principal component analysis (PCA) to property data, increasing user segmentation accuracy by 28.3%

Baidu, Inc.

Beijing, China

Senior Data Analyst | Baidu Healthcare & Baidu Map

Jul. 2022 – Jan. 2024

- Optimized marketing spend ROI by implementing SQL-based analytics framework; analyzed 2M+ ad performance metrics and applied regression models, reducing cost-per-mile by 64% while maintaining key conversion KPIs
- Drove acquisition of 8M+ new users in 6 months by developing user acquisition prediction system; engineered features and optimized hyperparameters to identify high-value market segments
- Implemented healthcare data quality framework with referential integrity checks and completeness validation; designed automated ETL workflows and anomaly detection processes, reducing record inconsistencies by 85%
- Automated media performance tracking with BI dashboards, to replace Excel Pivot Table manual updates, allowing budget reallocations every 6-12 hours, and resulted in reducing wasted ad spend by 20%, equivalent in \$550k

Digital China Group Co, Ltd

Beijing, China

Intern Analyst

Apr. 2022 – Jun. 2022

- Created industry-specific white papers and promotional content for smart city and IoT solutions, driving 10% more client engagement across Digital China's enterprise product line, from 200 to 220
- Developed performance reports and visualizations for 200+ enterprise clients; transformed complex IoT product data into actionable insights, resulting in 15% higher lead conversion rates for B2B marketing initiatives

Ogilvy & Mather Advertising Co, Ltd.

Beijing, China

Intern

Dec. 2021 – Mar. 2022

- Refined customer segmentation strategy for client's brand by building cluster models in Python, enabling 20+ targeted campaigns and boosting brand relevance scores by 12% (pre-test vs. post-airing, Kantar metrics)

PROJECT EXPERIENCE

Automated BI System for Bitcoin Price (Python, AWS Glue, Docker)

Sept. 2024

- Engineered serverless ETL pipeline using AWS Glue to extract real-time Bitcoin price data from CoinCap API; implemented data cleansing and transformation with PySpark, reducing processing time by 50%
- Enhanced data storage by designing optimized PostgreSQL schemas with efficient indexing on timestamp and price columns, enabling real-time price analytics queries and improving dashboard response times by 65%
- Implemented Docker containerization for consistent environment management across development and production, creating a portable system architecture that eliminated configuration issues and enabled seamless deployment